

Technical Document #592: Natural Language Processing - Comprehensive Overview

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Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Tokenization is the process of breaking text into individual units called tokens. Subword tokenization methods like BPE handle out-of-vocabulary words gracefully.

Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.

The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys. It is the core component of transformer models.

Language models predict the probability of a sequence of words. Large language models like GPT and BERT have achieved remarkable performance on various NLP tasks.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Key Takeaways:

- The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys.
- Language models predict the probability of a sequence of words.
- Coreference resolution identifies all expressions that refer to the same entity in a text.